Wittgenstein on Axioms in Lectures on the Foundations of Mathematics

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1. *Introduction*

The concept of an axiom, a fundamental component of mathematics, has remained unclear since its introduction in antiquity, when it was used to lay the groundwork for the field. Axioms have remained integral to mathematics over the centuries, yet our understanding of the concept has remained fretfully unclear. We generally define an axiom as a proposition that is assumed to be true without proof, either due to its apparent self-evidence or its ability to provide a stable basis for further argumentation. But this is not a satisfactory definition, it's the result of a long, fruitless search for something more adequate sparked by certain advancements in mathematics that rendered the previous one untenable. Up until the 19th century, it was thought that axioms were simply self-evidently true, so obvious that they need to be questioned. Such was the case until an attempt was made to reduce the whole of mathematics to a set of axioms that seemed self-evidently true but would turn out to lead to serious contradictions, forcing mathematicians to reconsider the view of axioms as simply obvious. Since then, mathematicians have appeased themselves with our current definition, which has addressed these issues but left us philosophically unsatisfied. After all, what compelling reason is there as to why axioms should not be self-evident? Despite numerous attempts to answer such questions, no single approach has prevailed

over the others and the notion of an axiom has remained vague because of it. Of these various attempts at clarifying the nature of axioms, a uniquely inventive and characteristically polemic perspective was proposed by Ludwig Wittgenstein, who intensely explored such questions for much of his life. Although he's most well-known for his philosophy of language, the subject of most of Wittgenstein's writings was in fact the philosophy of mathematics, an aspect of his work that has not been properly appreciated because he never actually published any of it. It was only with a set of lectures that Wittgenstein gave in 1939 that his ideas on the foundations of mathematics were cohesively laid out and made accessible to a wide audience. Although axioms were not the central focus of these lectures, it would have been impossible for Wittgenstein to discuss the foundations of mathematics without addressing the question of axioms, and he did so in his characteristic approach. Rather than proposing a novel conception of axioms, Wittgenstein focuses on highlighting the flaws in our ways of thinking about them and systematically disassembles the prevailing views on their nature. This essay will explore Wittgenstein's views on axioms, proofs, the objectivity of axioms, and the problem of contradictions as they are laid out in the published version of these lectures. By examining these aspects of his philosophy we'll come to see how Wittgenstein clears up our muddled conception of axioms, showing us that we were foolish for asking such questions in the first place.

2. *Axioms and proofs*

All mathematical arguments start somewhere; mathematics consists in deriving some propositions from other more primitive propositions which we call axioms. As justification that one proposition follows from the other, we provide a proof, one which is capable of showing how such a thing can be done. But what does it mean to show such a thing? Wittgenstein contends that there is nothing about the proof in and of itself to which we can appeal to to make others accept it. Asserting that a given proposition follows from some axioms merely states that such-and-such proposition follows from some other such-and-such more primitive proposition. This, on its own, does not seem like a convincing proof. Wittgenstein lays out his argument for why with an example:

> "...one can show a child how to multiply 24 by 37, and 52 by 96, and then say to it, "Now multiply 113 by 44 analogously." The child may then do one of many things. If he can't justify his action, we should go through it again and again, until we converted him to doing the same as us. The only criterion for his multiplying 113 by 44 in a way analogous to the examples is his doing it in the way in which all of us, who have been trained in a certain way, would do it. If we find that he cannot be trained to do it the same as us, then we give him up as hopeless and say he is a lunatic."

Here Wittgenstein discusses the process of teaching a child to multiply various numbers and then testing their comprehension of the process. He posits that the child's only criterion of success in multiplication is whether they can perform it in the same way in which they were taught. If the child cannot be trained to follow the same procedure, they are simply deemed unable to comprehend the concept. All we can appeal to so the child understands why his answer is incorrect is telling him that what he's doing is not the same as what he's been told to do and to try again. Wittgenstein uses this example to emphasize that accepting mathematical proofs doesn't rely solely on the formal structure of axioms and proofs. Rather, the correctness of the multiplication is rooted in the normative aspects of teaching, where the child is taught to regard certain mathematical truths as undeniable, leading them to accept the consequences of these truths. From this perspective, axioms can be understood as more akin to the rules of a game, a parallel to his concept of 'language games' in his philosophy of language. They are not necessarily self-evidently true but are practical inventions designed to help us navigate the mathematical landscape. Axioms do not inherently give meaning to mathematical propositions; instead, the true proof that a proposition follows from another lies in the actual act of transforming one into the other. In this context, the foundation of mathematics is not rooted in axioms but in the very act of engaging in mathematical thought and problem-solving. This is why the correctness of a child's

multiplication must be demonstrated through the actual act of multiplying—no other proof will suffice.

3. *The objectivity of axioms*

Wittgenstein's view of axioms seems intuitively wrong to anyone who has ever engaged in even the simplest of arithmetic calculations. As we work to add or subtract such-and-such numbers we get the sense that some operations are naturally more correct than others. This compels us to think of axioms as necessarily objective, as expressing some reality that is common to everyone. However, Wittgenstein asserts that this sense of correctness is not because axioms are objective rules that become self-evident to anyone engaging in mathematics. Axioms, according to him, cannot be verified experientially and do not make objective claims. To illustrate his point, Wittgenstein refers to one of Euclid's axioms:

> "Take the axiom: Between any two points, a straight line can be drawn. This is obviously untrue, you might say. Try to draw a line between a point on the moon and a point on Sirius, whatever a point may be. —But clearly that is not what is meant by 'can be drawn.'"

Here he demonstrates the difficulty in defining what an axiom really postulates, be it objective or not. For Euclid's axiom it is unclear what is actually being said 'can be done,' as we often end up with experiential propositions that are evidently inaccurate

when we try to form an answer. According to Wittgenstein, axioms exist apart from our empirical reality, not in a Platonic sense, but rather in the same way that the rules of a game exist. Consequently, axioms cannot be considered 'objective' in the traditional sense, as there is no common reference point apart from the process of doing the mathematical procedure itself. This addresses a pressing issue with Wittgenstein's view of mathematics as something like a game'; it doesn't matter that axioms aren't 'objective,' as they serve a very useful purpose. Wittgenstein doesn't believe mathematics is fake; rather, it is an invention that serves an incredibly useful purpose. By freeing mathematics from the strict need to be objective, his approach is more accurate to what actually happens in the field. In this sense, mathematics is like the rest of our language—a tool that helps us navigate and understand the world around us.

4. *The problem of contradictions*

A prevalent concern among mathematicians is the possibility that axioms, if not self-evidently true but rather invented for practical purposes, may be formulated inaccurately and potentially lead to contradictions somewhere down the line. This concern is a principal motivation for an idealistic conception of axioms as self-evident, which is rooted in our intuitive urge to seek objective, common ground when assessing their merit. Wittgenstein once again makes us question our intuitions by contending that the conception of inconsistencies as problematic is not well-founded in the first

place. He contends that contradictions do not actually pose a significant problem for mathematical systems and that there's no reason to think that a single inconsistency should cause the entire system to crumble. To illustrate his point, Wittgenstein uses the following example:

> "Suppose we now use our rules, and one day we arrive at a contradiction. We may then say that we have not used the rules correctly; or we may want to change the rules. I may give you the rules for moving chessmen without saying that you have to stop at the edge of the chessboard. If the case arises that a man wishes to make a piece jump off the chessboard, we can then say, "No, that is not allowed." But this does not mean that the rules were either false or incomplete. —Remember what was said about counting. Just as one has freedom to continue counting as one likes, so one can interpret the rule in such a way that one may jump off the board or in such a way that one may not. But it is vitally important to see that a contradiction is not a germ which shows general illness."

Wittgenstein highlights how contradictions are not always indicative of a fundamental flaw in our axiomatic systems, emphasizing that contradictions can be easily managed and should not be considered inherently destructive. Just as we have the freedom to continue counting in various ways, we can also interpret rules in different ways. The presence of a contradiction does not necessarily imply that the entire system is invalid

or diseased; it simply indicates that we may need to reconsider or clarify our rules. Wittgenstein's view challenges the traditional belief that contradictions are inherently problematic and instead suggests that they can be addressed and resolved within the system. In a manner similar to the game of chess, where one might incorrectly interpret the rules and move a piece off the board, we can encounter a contradiction in an otherwise functional mathematical system without becoming distressed by the possibility of such an occurrence. Instead of panicking, we simply work through the issue, clarifying the rules and refining our understanding until we arrive at a satisfactory resolution. This approach allows us to navigate mathematical challenges with the flexibility and adaptability of which they are capable, ultimately leading to a more robust understanding of the concepts at hand.

5. *Conclusion*

Wittgenstein's view of axioms can succinctly be summed up in his response to a remark by Bertrand Russell:

> "Russell says, roughly: "After all, it is not self-evidence which must guide one in the choice of primitive propositions. On the contrary, one is guided sometimes by the results which a given choice produces. Many primitive propositions are shown to be true by what follows from them."

—You may choose them because you want to get to a certain point. Not because they are indubitable."

Rather than attempting to disprove the mainstream conception of mathematics, Wittgenstein encourages us to question it, exposing the uncertainty underlying our previously held beliefs. In doing so, he brings a sense of redemption to the initially unsatisfactory definition of an axiom which we laid out in the outset. Though it lacks the certainty we typically desire, Wittgenstein's conception of mathematics embraces this fluidity. In his view, contradictions do not signify the end of our systems; instead, they are part of their constant evolution as a game we engage in. Wittgenstein emphasizes the flexible nature of axioms, considering them to be whatever we need them to be in order to achieve our desired outcomes, functioning like the rules of a game we've created. His approach highlights that mathematics, much like language, is a tool we use to navigate and understand the world around us. Therefore, our understanding of its foundations need not be rooted in some unattainable, objective reality. By appreciating the value and utility of mathematics as an inventive and immensely useful construct, we can recognize axioms as guiding principles that facilitate our engagement in mathematical thought and problem-solving. This perspective allows us to better appreciate the dynamic nature of mathematics as a human invention, rather than a rigid system based on objective truth.

Bibliography

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